GroEL CXIDB ID 187 External Processing Checklist

Goal

- Do not store the full CXIDB raw data on this PC.
- Use an external machine with enough storage to download and process the raw data.
- Bring back only the processed assembled hits and, if available, the photon background.
- The first validation target should be run 151, because the Ekeberg analysis used run 151 for the representative single-pattern matching example.

First-pass scope

- Data run: r0151.
- Dark run: r0149, following the DATA_RUNS/DARK_RUNS mapping in the Ekeberg repository.
- Export only the strongest 100-500 hits at first. The full hit set is not required for the initial validation.
- If possible, preserve the mapping from exported hit index back to the original frame index.

Non-negotiable constraints

- Do not use private coned-deformed droplet data.
- Use only public CXIDB/Ekeberg GroEL data for this validation section.
- Do not copy the full raw PNCCD run back to this PC.

Data and Environment on the External Machine

Required raw/support files

- CXIDB ID 187 raw PNCCD files for r0151.
- Raw PNCCD files for dark run r0149.
- histograms.h5, mask.h5, rotations.h5, 1SS8.pdb, and 1SS8_H2O.pdb.
- The support files already exist on this PC, but the external machine also needs them for processing.

Repository

- Use https://github.com/ekeberg/Ekeberg2022GroEL as the processing reference.
- Set the XFEL2146_DIR environment variable to the Ekeberg2022GroEL root directory.
- The repository scripts assume a Data/ and Results/ folder layout under XFEL2146_DIR.
- Some scripts contain author-local absolute paths. Update them before running.

Recommended folder layout

```
- Ekeberg2022GroEL/Data/r0151/  -> run 151 raw PNCCD files.
- Ekeberg2022GroEL/Data/r0149/  -> dark run 149 raw PNCCD files.
- Ekeberg2022GroEL/Data/histograms.h5.
- Ekeberg2022GroEL/Data/mask.h5.
- Ekeberg2022GroEL/Results/     -> all processed outputs.
```

External Processing Steps

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1. Fix paths and set environment

- Set XFEL2146_DIR to the Ekeberg2022GroEL directory.
- In Scripts/fit_gain_parameters.py, point histograms_file to Data/histograms.h5.
- In Scripts/xfel2146_tools.py, make mask() read Data/mask.h5 instead of the original author-local path.

2. Dark and gain calibration

- Run prepare_dark.py for dark run r0149.
- Run fit_gain_parameters.py and make_gain_map.py to produce Results/Gain/gain_map.h5.
- If the external PC does not use SLURM, replace array-job logic with a loop over detector lines 0-1023.

3. Hit extraction

- Run count_lit_pixels.py for r0151.
- Run export_hits.py to convert selected hits into photon-count patterns.
- Run assemble_hits.py to produce assembled 1024 x 1024 hit patterns.

4. Background

- If practical, run photon_background.py and assemble_background.py for the same detector geometry.
- The local retrieval script can run without a background file, using a constant background term.
- A real photon background is preferred because it is closer to the Ekeberg matching model.

Minimum Files to Bring Back to This PC

Required file

- `Results/AssembledHits/hits_r0151.h5`.

- Required dataset: patterns_ph, shape (n_hits, 1024, 1024), photon-count hits.

- Required dataset: mask, shape (1024, 1024), valid detector mask.

- Optional but useful dataset: lit_pixels, shape (n_hits,).

Recommended file

- `Results/Background/photon_background_r0151.h5`.

- Recommended dataset: photon_background, shape (1024, 1024).

Storage reduction

- For the first transfer, keep only the strongest 100-500 hits.

- Store patterns_ph as uint16 or float32 if possible.

- Use HDF5 lzf or gzip compression.

- Do not transfer raw PNCCD files, dark raw files, or full intermediate corrected runs.

Where to Put Files and What to Run Locally

Copy target on this PC

```
- C:\Users\tnvlaos456\Documents\SAXS
  simulation\public_data\CXIDB_ID_187\Results\AssembledHits\hits_r0151.h5
- C:\Users\tnvlaos456\Documents\SAXS
  simulation\public_data\CXIDB_ID_187\Results\Background\photon_background_r0151.h5
```

Build a small template library first

```
- .\.venv_cxidb\Scripts\python.exe
  validation\section5_groel_generate_template_library.py --models 8 --include-prism-
  models --prism-radii-A 35,55 --rotation-start 0 --rotation-stop 100 --output-dir
  validation\groel_template_library
```

- Start with about 100 rotations as a smoke test. Increase the rotation range only after this works.

Run retrieval

```
- .\.venv_cxidb\Scripts\python.exe validation\section5_groel_retrieval.py --pattern-
  file public_data\CXIDB_ID_187\Results\AssembledHits\hits_r0151.h5 --pattern-dataset
  patterns_ph --mask-dataset mask --background-file
  public_data\CXIDB_ID_187\Results\Background\photon_background_r0151.h5 --background-
  dataset photon_background --template-files validation\groel_template_library\*.h5
  --output-dir validation\groel_retrieval_results --top-pattern-count 5 --top-k 10
  --shift-radius 6
```

Final Checklist

On the external machine

- [ ] r0151 raw data downloaded.
- [ ] r0149 dark run downloaded.
- [ ] hard-coded histograms.h5 and mask.h5 paths fixed.
- [ ] gain_map.h5 generated.
- [ ] hits_r0151.h5 contains patterns_ph and mask.
- [ ] photon_background_r0151.h5 generated, or the no-background choice is documented.
- [ ] exported hit index to original frame index mapping is preserved if hits were filtered.

After copying files to this PC

- [ ] hits_r0151.h5 placed under public_data/CXIDB_ID_187/Results/AssembledHits/.
- [ ] photon_background_r0151.h5 placed under public_data/CXIDB_ID_187/Results/Background/ if available.
- [ ] a small template-library smoke test was run.
- [ ] retrieval CSV and PNG outputs were checked.